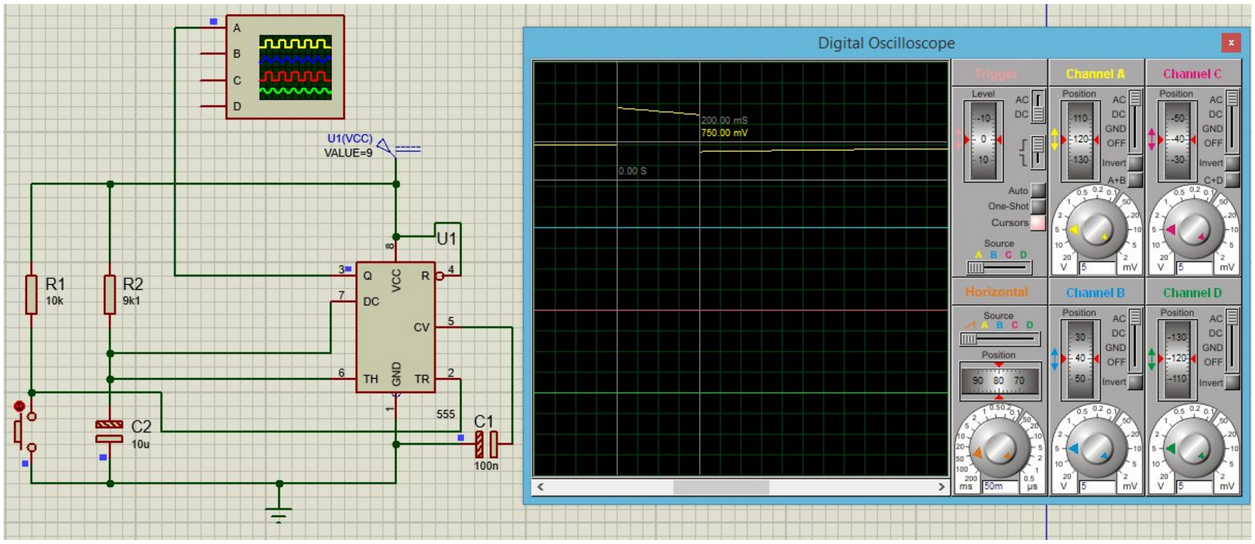
Thejasvenan T. 220638J

ME 2261 - Embedded Systems Log Book

Day 02 – 25.07.2025

Lab Practical 2

Task 1



T = 1.1RC

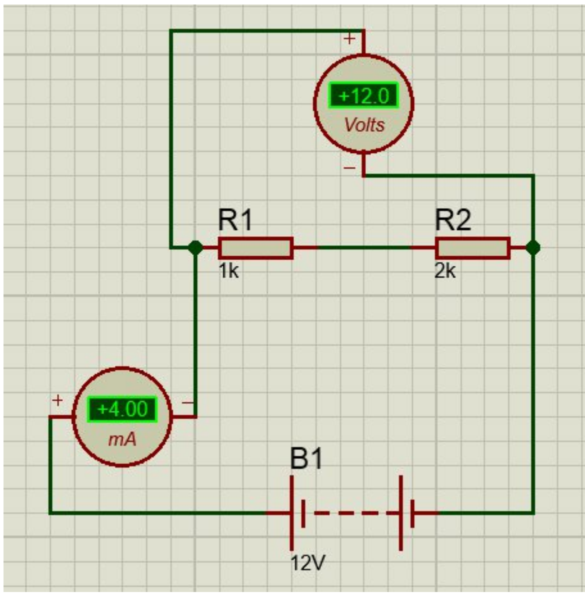
= 1.1 x 9.1 kΩ x 10 uF

= 100.1 ms

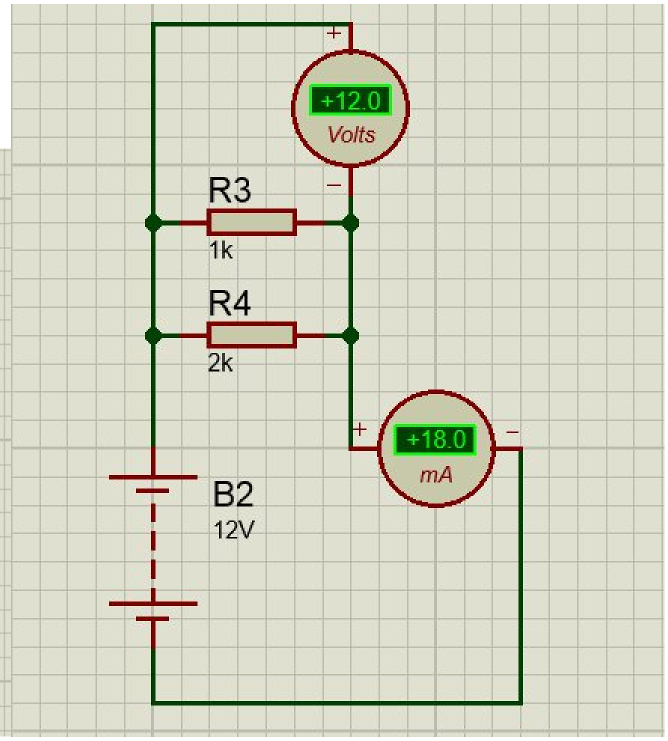
Time period obtained in the Oscilloscope = 200 ms

Task 2

Series



Parallel



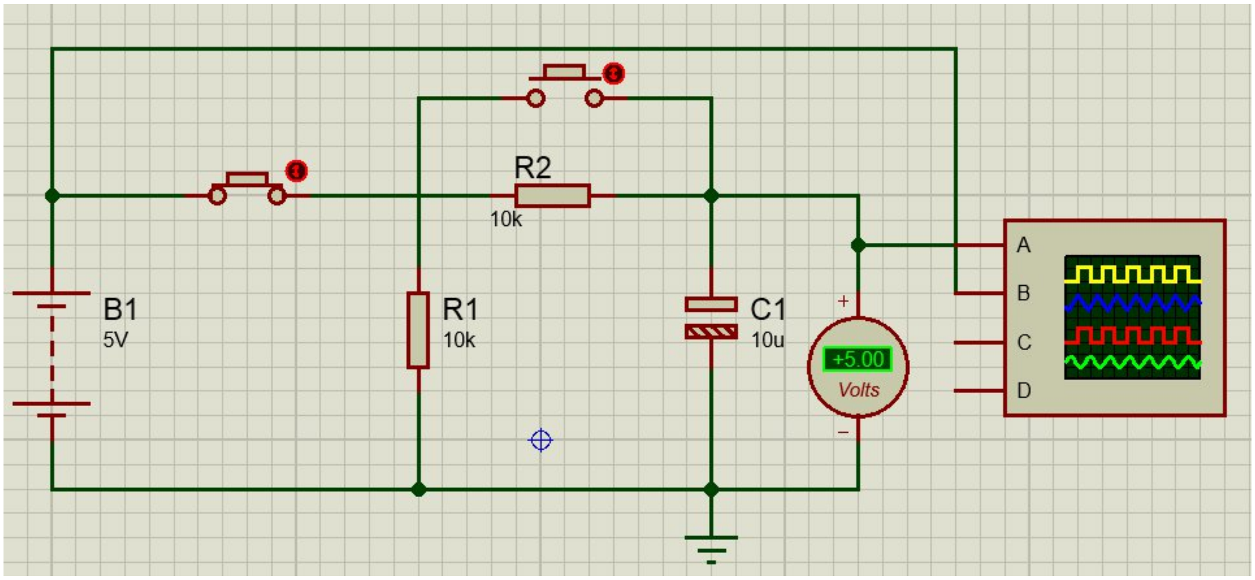
Total Resistance = V/I

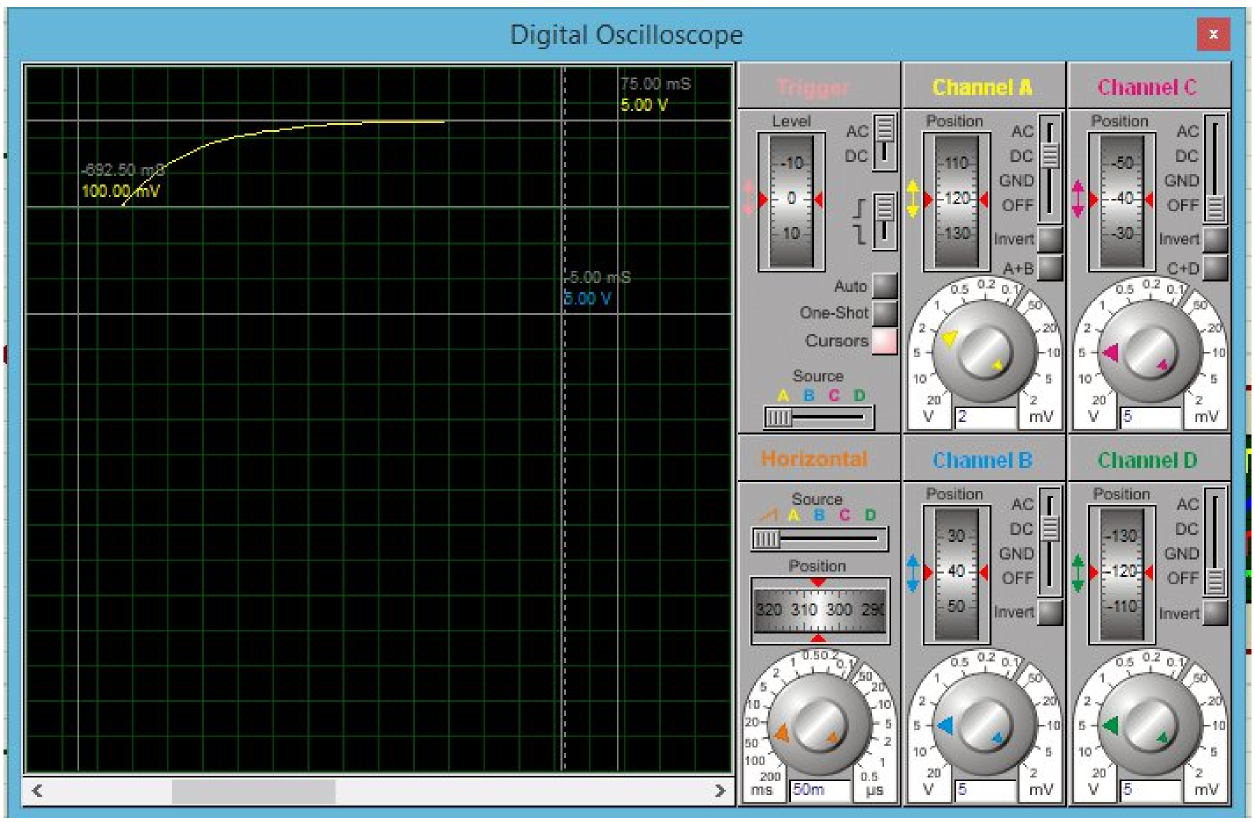
In Series mode, TR = 12/4 x 10 = 3 kΩ

In Parallel mode, TR = 12/18 = 0.66 kΩ

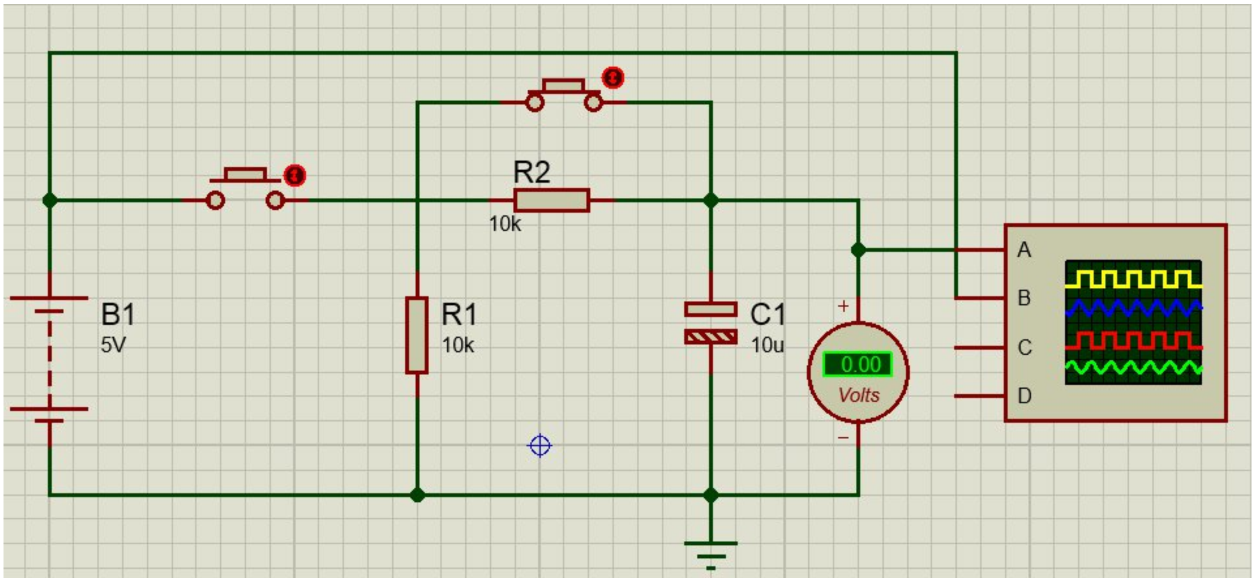
Task 3

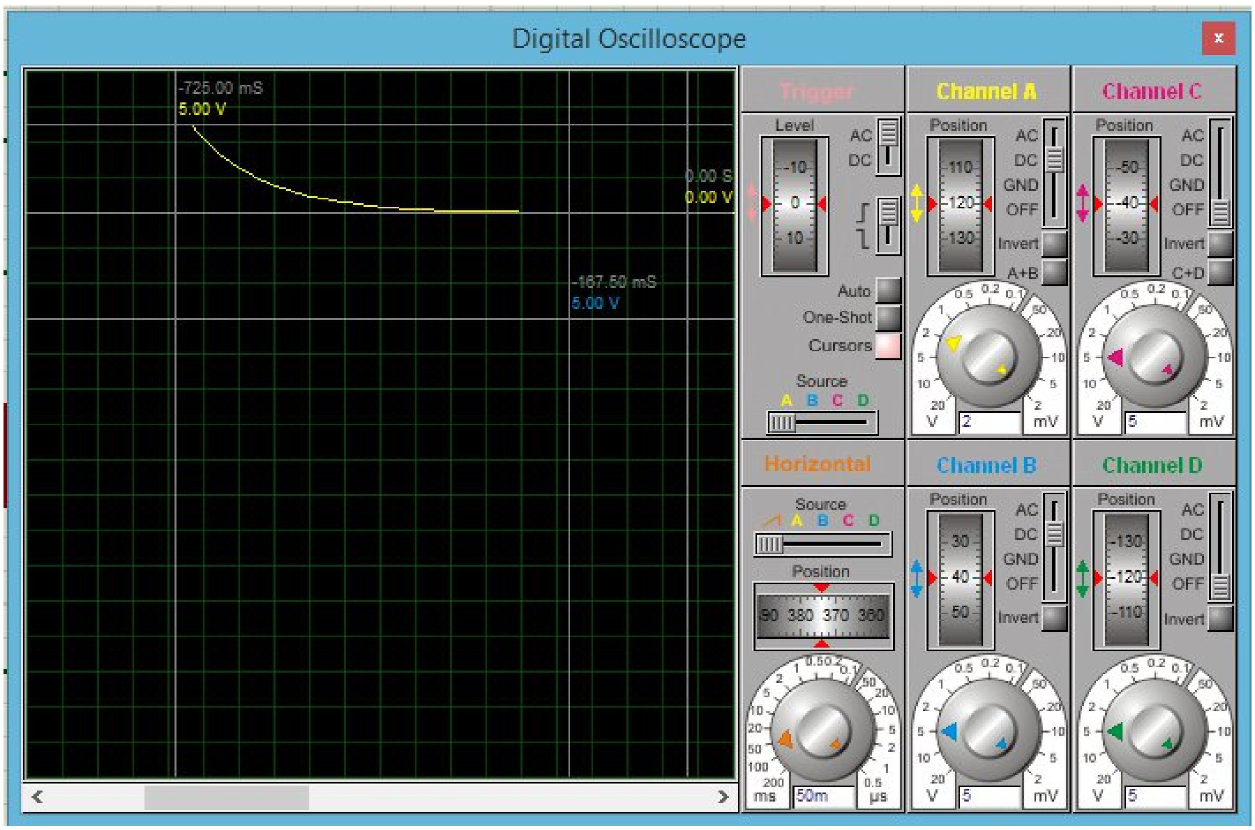
Charging process





Discharging process



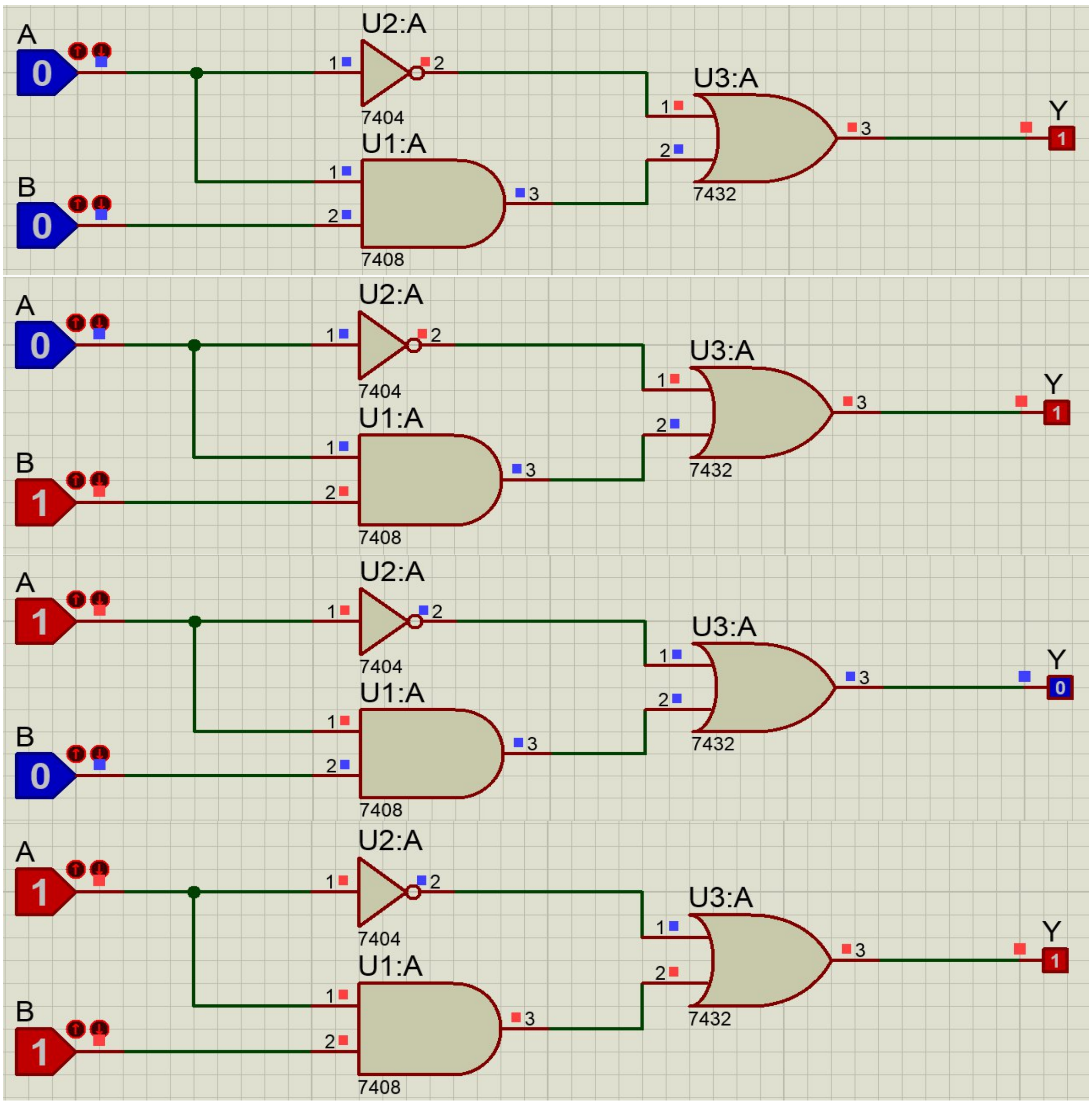


Task 4

Truth table

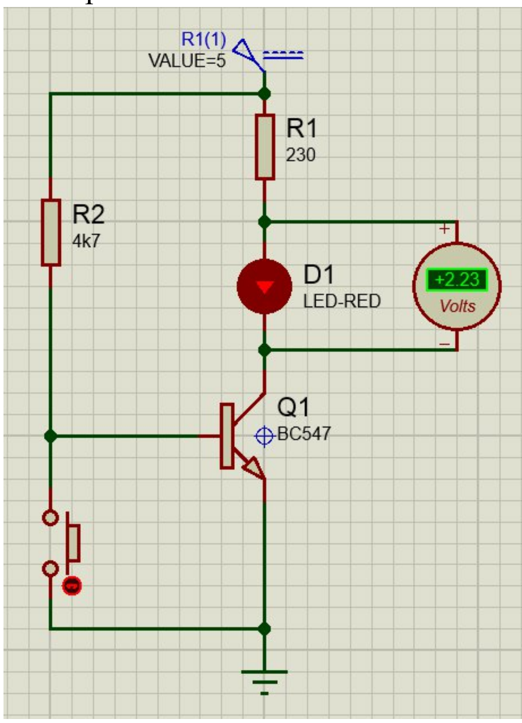
$$Y = (A \cdot B) + \overline{A}$$

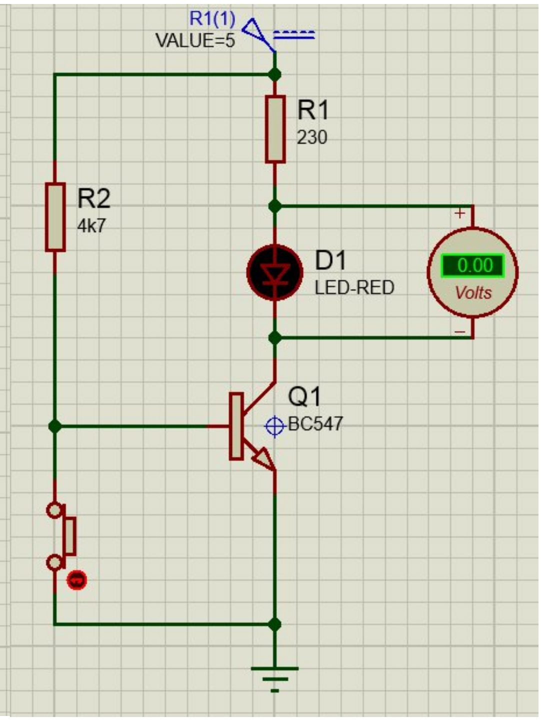
A	**B**	$A \cdot B$	$\overline{A}$	$(A \cdot B) + \overline{A}$
0	0	0	1	1
0	1	0	1	1
1	0	0	0	0
1	1	1	0	1



<u>Task 5</u>

Pull up Resistor

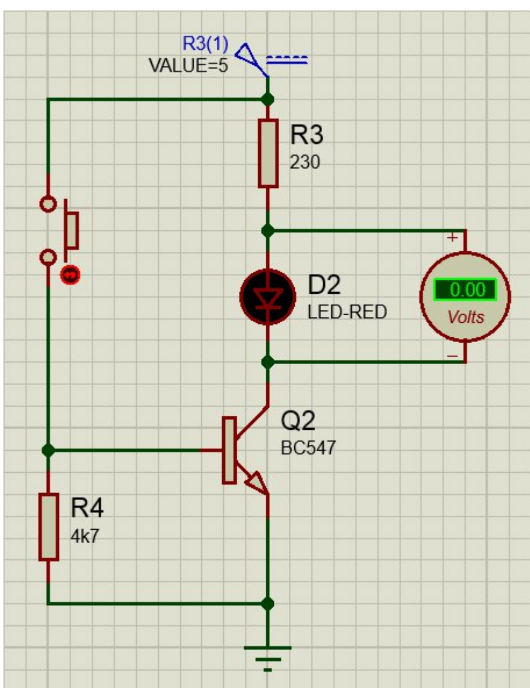


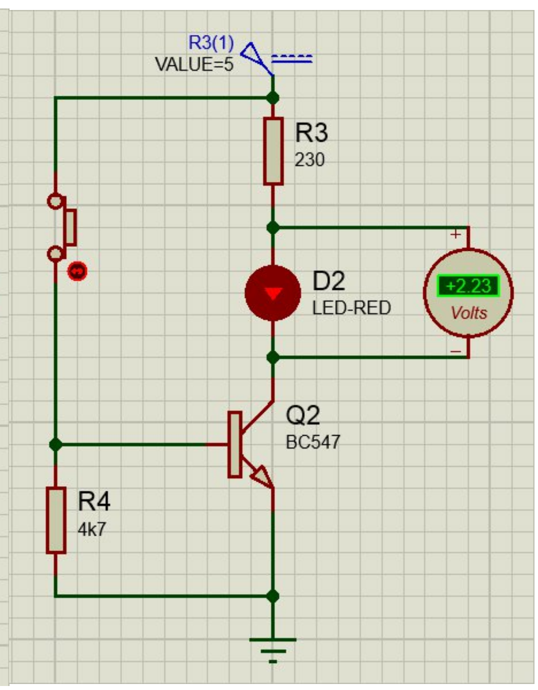


Input is LOW

Input is HIGH

Pull Down Resistor





Input is LOW

Input is HIGH